

**TO:** Dr. Azade Farshad & Doctoral Selection Committee · Department of Computer Science, Aalto University / ELLIS Finland

**SUBJECT:** Application for Doctoral Studies (Autumn 2026) — Bidirectional Foundation Architectures for Continuous Sign AI

**COMPANION DOSSIER:** Doctoral Proposal: [Proposal.pdf](#) | Kinematics Engine: [Live WebGL Demo](#) | Code: [GitHub](#)

Dear Dr. Farshad and Members of the Doctoral Selection Committee,

I am writing to express my enthusiastic interest in pursuing doctoral research in the Department of Computer Science at Aalto University and the ELLIS Institute Finland under the supervision of **Dr. Azade Farshad**, directly advancing ELLIS Finland's strategic agenda in *AI for Health & Assistive Digital Twins*.

## 1 Part I: Why I Fit the Role — Motivation, Linguistic Foundations, and Lab Synergy

### The Catalyst: An Unsolved Computational Frontier & Healthcare Imperative

My drive to pursue doctoral research is rooted in a profound intellectual fascination with **complex, fundamentally unsolved mathematical and computational problems**. This exact challenge drew me to quantitative finance and continuous reinforcement learning during my Master of Science research at NIELIT Calicut (Project Bithax)—investigating non-stationary distributions, continuous policy optimization, and stability bounds under noisy constraints. Yet, exploring sign language artificial intelligence revealed an even more compelling, open scientific frontier: unlike spoken language, signing is an unwritten, multi-channel 4D continuous phenomenon where discrete semantic meaning is inextricably bound to musculoskeletal biomechanics, simultaneous articulatory co-trajectories, and 3D spatial geometry.

This theoretical challenge crystallized into an urgent research calling during an emergency hospital triage visit when a close friend required blood donation. In the clinic, I assisted a migrant patient unable to describe acute symptoms to medical staff due to a severe language barrier; translating between regional languages, I witnessed firsthand how survival and clinical outcomes depend on mutual intelligibility. Later encountering Deaf signers and discovering the complete absence of real-time computational translation systems, the gravity of this technological void became undeniable: over 70 million Deaf individuals face systemic communicative exclusion. Under the **World Health Organization (WHO)** mandate defining health as complete physical, mental, and social well-being, and Finland's statutory framework under **Finland's Sign Language Act (Viittomakielilaki 359/2015)** and the European Accessibility Act (Directive 2019/882), my mission is to address this unsolved frontier through an autonomous, bidirectional continuous sign foundation system.

### Sign Language Taxonomy, Linguistic Lexicons & Literature Gaps

Sign languages cannot be modeled as simple text translations: **they possess no conventional written orthography** and communicate grammar through continuous spatio-temporal dynamics across 4D physical space. In formal sign linguistics (Stokoe, 1960; Battison, 1978), every sign is characterized by five sub-lexical parameters:  $S = \langle H, L, M, O, F_{NMS} \rangle$  (Handshape, 3D anatomical Location, Movement trajectory, Palm Orientation, and Non-Manual Signals including facial Action Units and head tilt). Linguistic repositories, notably **ASL-LEX** and **ASL-LEX 2.0**, systematically catalog thousands of signs with quantified phonological distributions, continuous spatial coordinates, and lexical frequencies.

**Phonological Minimal Pairs & Referential Loci (*R-loci*):** Centimeter-scale positional deviations completely invert semantic meaning: in ASL, **MOTHER** (chin contact) versus **FATHER** (forehead contact) requires sub-centimeter vertical accuracy to avoid inverting maternal and paternal kinship. Similarly, **FATHER** (single tap) versus **GRANDFATHER** (double-arc outward bounce) is distinguished by movement velocity, while subtle palm orientation separates **NAME** from ungrammatical forms. Furthermore, signers project discourse entities onto virtual 3D coordinates termed **Referential Loci (*R-loci*)**, which directional agreement verbs (**INVITE**, **GIVE**) traverse across space. Existing Sign AI paradigms suffer from three fundamental theoretical bottlenecks:

- The Discrete Gloss Bottleneck in SignLLMs:** Multimodal models force 4D signing into 1D text tokens, discarding concurrent multi-channel co-articulation across dual hands, facial expressions, and spatial gaze.
- Unconstrained Diffusion & Articulatory Smearing:** Denoising mesh vertices without kinematic manifold constraints causes joint dislocations and “rubber finger” blurring that smear delicate minimal pairs.
- Absence of 3D Referential Spatial Memory:** Sliding 2D window architectures fail to maintain persistent 3D discourse anchors (*R-loci*), triggering catastrophic memory collapse on inflected agreement verbs.

### Technical Foundation & Supervisor Synergy: Why Dr. Azade Farshad's Lab

Under Dr. Vimala Mathew at NIELIT Calicut, my M.Sc. thesis in stochastic optimal control (Project Bithax) investigated continuous state-space MDPs and stability bounds under PPO, rigorously stress-testing systems against look-ahead bias. In parallel, through independent research mentorship with Dr. Marco Raoul Marini (Sapienza University of Rome), I investigated SLT/SLP formalization, sub-lexical decomposition, and ASL-LEX 2.0 phonological geometries.

Applying to **Dr. Azade Farshad's laboratory** is driven by deep theoretical convergence. Overcoming generative diffusion artifacts directly requires Dr. Farshad's pioneering work on topological regularizers and anatomical graphs (*SCOPE*, *HieraSurg*) to formulate a Riemannian manifold projection operator  $\Pi_{\mathcal{M}}$  guaranteeing avatars respect skeletal bounds. Furthermore, grounding dynamic spatial discourse anchors directly inherits Dr. Farshad's 3D semantic scene-graph methodology (*SceneGenie*), scaled on CSC's **EuroHPC LUMI supercomputer**.

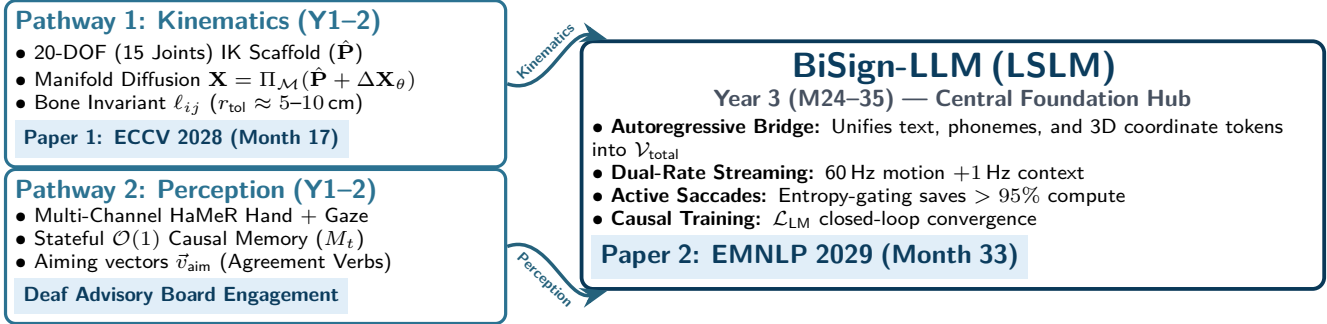


Figure 1: BiSign Foundation Architecture: Kinematic and Perceptual Pathways Converging into the Central LLM Hub.

### Closed-Loop Architecture & Demonstrated Feasibility Baseline

Rather than treating sign language as isolated video classification or ungrounded video generation, our framework establishes a closed-loop conversational architecture where perceptual understanding and physical synthesis reinforce each other:

- **Perceptual Understanding with Persistent 3D Spatial Memory:** When a Deaf user signs into a monocular camera, our perception pipeline tracks dual-hand articulations, facial blendshapes, and gaze. Crucially, an external  $\mathcal{O}(1)$  Causal Spatial Memory Bank preserves persistent 3D coordinate anchors ( $R$ -loci) across discourse turns, resolving inflected agreement verbs (INVITE, GIVE) without sliding-window memory loss.
- **Multimodal Foundation Core as Linguistic Bridge:** We adapt an open-weight foundation model on EuroHPC LUMI via vocabulary expansion ( $\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$ ), treating sub-lexical parameters and spatial targets as native linguistic tokens alongside text, capturing true spatial grammar rather than 1D gloss linearizations.
- **Anatomically Grounded Kinematic Synthesis:** In reverse translation, predicted sign tokens are compiled into continuous 3D motion projected onto an explicit human musculoskeletal manifold, eliminating “rubber finger” blurring and preserving delicate phonological minimal pairs.

To verify that these physical constraints operate with sub-millisecond responsiveness before training neural models, I developed an interactive **Dual-Hand Procedural WebGL Engine** ([Live WebGL Demo](#)). Formulating **5 foundational parametric lexicons** (41 handshapes, 24 loci, 10 orientations, 16 movements, 4 bilateral modes) and a **universal generator** spanning all 2,723 signs in ASL-LEX 2.0, the engine executes 30 articulated phalangeal joints at **60 FPS** in-browser (< 1 KB/s transfer). Integrating a 7-point volumetric soft-body repulsion barrier and quartic strain minimization  $P_{\text{strain}}(\theta)$ , automated audits achieve **100% PASS with 0 inter-finger self-collisions** across all 26 alphabet letters and 50 benchmark signs, empirically proving that explicit geometric grounding introduces zero latency overhead.

### Four-Year Doctoral Research Work Plan (Parallel Tracks & Staggered Milestones)

Synchronized with our foundation dossier and Aalto University’s cumulative doctoral guidelines:

- **Years 1–2 (M1–M16) — Parallel Foundational Modules (Track A & Track B in Parallel):**
  - *WP1 (M1–M16, Track A: Biomechanical Actuator & Kinematic Scaffolding):* Implement dual-hand kinematics with Battison phonological laws and closed-form arm IK with humeral swivel. Deploy manifold-projected residual diffusion ( $\Pi_{\mathcal{M}}$ ) calibrated against motion capture. **Paper 1 (M17, Target: ECCV 2028).**
  - *WP2 (M1–M16, Track B: Perceptual Tokenizer & Causal Memory — Parallel):* Harvest in-the-wild signing videos, deploy monocular hand tracking and facial blendshapes, and build the decoupled  $\mathcal{O}(1)$  Causal Spatial Memory Bank ( $M_t$ ) with deictic parity inversion  $\mathbf{R}_y(\pi)$  and *SceneGenie* grounding.
- **Year 3 (M17–M36) — Foundation Model Adaptation & LUMI Pre-Training (Track C):**
  - *WP3 (M17–M26):* Dual-rate streaming and vocabulary expansion ( $\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$ ) on open-weight foundation models.
  - *WP4 (M24–M32):* Continual pre-training on EuroHPC LUMI (AMD MI250X nodes, PyTorch FSDP + ROCm) across harvested sign tokens. **Paper 2 (M33, Target: EMNLP 2029).**
  - *WP5 (M31–M36):* Task-grounded downstream evaluation (Argument F1, minimal-pair preservation). **Paper 3 (M36, Target: ICLR 2030).**
- **Year 4 (M37–M48) — Clinical Pilot, Statutory Pre-Examination & Defense:**
  - *WP6 (M37–M41):* Pilot deployment in Finnish healthcare triage in participatory co-design with the Finnish Association of the Deaf (*Kuurojen Liitto ry*) under *Viittomakielilaki 359/2015*; open-source foundation release.
  - *Pre-Examination & Defense (M42–M48):* Cumulative dissertation compilation, formal pre-examination (*esitarkastus*) by external opponents, and public doctoral defense (*väitöstilaisuus*, Month 48) at Aalto University.

### Research Commitment & Methodological Agility

I understand that a PhD is fundamentally a 4-year intensive research journey requiring deep theoretical grounding, empirical discipline, and methodological agility. Because our architecture decouples perception, linguistic reasoning, and biomechanics, emerging foundational breakthroughs over the coming years—such as self-supervised spatio-temporal representations (e.g., V-JEPA) and ultra-efficient quantized open-weight LLMs—will be actively integrated to improve, adapt, and accelerate our research goals without compromising core geometric safeguards. I respectfully invite the doctoral selection committee to review our **Doctoral Research Proposal** ([Proposal.pdf](#)) for detailed sign language grammar and taxonomy, literature review, and complete breakdown of methodology and research path.

Respectfully submitted,

**Iniyan Andrews Joseph**